

Brahmi Sarah Kaouthar

Cybersecurity Engineering Student · AI & ML Researcher · Full-Stack Developer

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PROFILE

Cybersecurity Engineering student (5th year, expected 2027) at École Supérieure d'Informatique, Sidi Bel Abbès. Passionate about merging offensive security, AI/ML research, and full-stack development to build smart and secure solutions. Team-lead experience on autonomous RL-based attack-defence simulations. Active community contributor via GDG and Alphabet clubs.

SKILLS

Cybersecurity	Kali Linux, Metasploit, Burp Suite, Wireshark, Nmap, Suricata, Hydra, John the Ripper, Hashcat, Nikto, Autopsy, Volatility, tcpdump, Zeek, ModSecurity
SIEM / Monitoring	Sysmon, Winlogbeat, Elasticsearch, Kibana, Telegraf, InfluxDB, Grafana, Prometheus
AI & Machine Learning	Python, BERT, UNET, EfficientNet-B3, OpenCV, Deep Q-Network, NLP, Haar Cascade
Frontend & Mobile	React.js, Flutter, Dart, HTML5, CSS3, JavaScript, Tailwind CSS, Vite
Backend & Databases	FastAPI, MySQL, MongoDB, Cassandra, Neo4j, Firebase, OLAP, NoSQL, PL/SQL
DevOps & Cloud	Docker, Kubernetes, Git, Jenkins, Ansible, LocalStack, GNS3, ZeroTier
Languages	Python, JavaScript, Dart, Java, C, Pascal, Bash
Design	Figma, Adobe Photoshop

EDUCATION

Cybersecurity Engineering

École Supérieure d'Informatique, Sidi Bel Abbès

2022 – Expected 2027

Baccalaureate in Mathematics

Algeria

2022

Average: 16.41/20

Baccalaureate in Sciences

Algeria

2021

Average: 16.2/20

TRAINING

Internship — Algérie Télécom

Network fundamentals and security, system administration, real-world telecom infrastructure.

EXTRACURRICULARS

Alphabit Club (2023–Present)

Marketing · Social media content · Event certificates

GDG Club (2022–2023)

Graphic design · Video editing · DevFest 2022 organisation

SOFT SKILLS

Leadership · Problem-solving · Creativity · Attention to detail · Adaptability · Communication · Teamwork

PROJECTS

Cyber-Arena — Autonomous Cybersecurity Simulation Platform [Team Lead]

GNS3 · Deep Q-Network · Telegraf → InfluxDB → Grafana · Ansible · ZeroTier

- Real-time simulation where Red (attacker) & Blue (defender) DQN agents learn on a full enterprise topology: 8 PCs,

5 VLANs, 2 Cisco routers, Alpine firewall.

- Implemented Cyber Kill Chain (Reconnaissance → Intrusion → Lateral Movement → Impact) with 80-dim state space and 18 actions per agent.
- Built live telemetry pipeline (SNMP → Telegraf → InfluxDB → Grafana) and honeypot system; Ansible automates daily config backups for 6 devices.

Ransomware Simulation & Detection

Sysmon - Winlogbeat - Suricata - Elasticsearch - Kibana

- Simulated ransomware behaviour and built endpoint monitoring pipeline to detect, alert, and visualise attack patterns in real time via ELK stack.

Serverless Function Simulator

Python - FastAPI - LocalStack - JSON Structured Logging

- Custom Python FaaS runtime with HTTP, SQS queue, and cron triggers; cold start simulation (350ms), warm container pool (0ms), and p50 latency telemetry.

Pink Diary - Encrypted Personal Journal App

React.js - Vite - Tailwind CSS - End-to-End Encryption

- Feature-rich diary app with advanced encryption, rich media entries, and polished UI/UX.

LTE-2G Downgrade Attack Simulation

Network Security - Protocol Analysis - Python

- Educational simulation of IMSI catcher-based LTE to 2G downgrade for security research.

AI Projects - Image Segmentation & NLP

Python - BERT - UNET - EfficientNet-B3 - OpenCV

- AISM Challenge: semantic similarity with BERT on Kaggle datasets.
- Image Segmentation: UNET + EfficientNet-B3 for accurate segmentation tasks.